Phantom Brand Persistence Has a Living-Brand Boundary

On a heritage-saturated automotive substrate with a dedicated Cell D panel of discontinued corporate brands, the AI mediation layer surfaces living heritage brands at elevated rates (R_phantom up to 14/18) but does not surface any of the five defunct brands a single time across 18 unprompted Recall opportunities each. The lead hypothesis falsifies cleanly; the supporting heritage hypothesis confirms cleanly. The phantom phenomenon has a temporal floor.

May 2026 · Pablo Ulpiano González Castro · Third System

Independent measurement for the AI mediation layer.

AIAS™ v0.23 stress-tests Phantom Brand Persistence on a heritage-saturated automotive substrate, with a dedicated Cell D panel of five discontinued corporate brands (Pontiac, Oldsmobile, Plymouth, Mercury, Saturn) providing the pure-phantom upper-bound test. The result inverts the program's working hypothesis: despite all five defunct brands clearing Phase A Recognition at C_P = 6/6, none surfaces once in any of 18 unprompted current-tense Recall opportunities per brand. Meanwhile, three Cell A heritage brands (Mercedes-Benz, Porsche, BMW) cross the R_phantom ≥ 8 CONFIRMED threshold. The phantom phenomenon operates on living heritage brands and stops at brand death — a tighter and more informative claim than "phantoms surface broadly."

Four findings shape the v0.23 phase. First, Recognition saturates uniformly across all four cells — 23 of 24 brands at C_P = 6/6, one at 5/6 — routing v1.5 C2 to failure in every cell and v1.6 Inc1 substrate Recognition Pre-Screen to UNIFORM SATURATION as predicted. Second, Iwachu-pattern dissociation generalizes to the program's sixth substrate family with cases in all four cells, though five of seventeen cases are Cell D defunct brands that meet the Iwachu criterion tautologically by the lead falsification. Third, the heritage substrate produces the largest single-phase Type 2 count in the program (seven cases) and exposes a clean within-cell channel bifurcation: Mercedes-Benz, Porsche, and BMW surface heavily in both canonical and cultural Recall, while Rolls-Royce, Bentley, Jaguar, and Cadillac surface only in cultural / heritage frames. Fourth — the lead finding — no defunct brand surfaces in any Phase B response, falsifying H_Phantom_Defunct at the strongest possible margin and establishing the temporal boundary of the phantom phenomenon.

AIAS™ measures AI Availability — the probability that an AI intermediary retrieves, recommends, or selects a brand in a category-anchored decision context — as a third measurable layer of brand availability alongside the Ehrenberg-Bass framework's Mental and Physical Availability. Protocol v1.6 (SSRN 6816340) introduced three increments: a substrate Recognition pre-screen distinguishing uniform from differential saturation (Inc1), a direct R4-independent IL measurement (Inc2), and an explicit Phantom Brand

Persistence measurement (Inc3). v0.23 is the first prospective phase under v1.6 lock, retaining v1.5's two-channel Recall decomposition (R_cat and R_cult) and v1.2's four-regime taxonomy.

The substrate is automotive, sampled across four cells of seven, five, seven, and five brands. Cell A (Heritage, n = 7): Mercedes-Benz, Jaguar, Cadillac, Rolls-Royce, Bentley, Porsche, BMW. Cell B (Disruptor, n = 5): Tesla, Rivian, Lucid, Polestar, Fisker. Cell C (Mass-Legacy, n = 7): Toyota, Honda, Ford, Chevrolet, Hyundai, Volkswagen, Nissan. Cell D (Defunct, n = 5) — the pure-phantom panel novel to v0.23: Pontiac (closed 2010), Oldsmobile (2004), Plymouth (2001), Mercury (2010), Saturn (2010). Reference panel: the six-slot LLM panel carried forward from v0.17 (Claude Opus 4.5, Claude Sonnet 4.5, GPT-4o, GPT-4o-mini, Gemini 2.5 Flash, Gemini 2.5 Flash Lite).

Six hypotheses were pre-registered (tags v0.23-prereg-r1 superseded by v0.23-prereg-r2 — the r1 → r2 amendment corrected an INSTRUMENT specification defect caught pre-acquisition; see Limitations). H_Phantom_Defunct, the lead hypothesis, returned FALSIFIED at the strongest possible margin: all five Cell D brands produced R_phantom_defunct = 0 across the six-model panel × three R_cat probes = 18 mention opportunities per brand. The supporting hypothesis H_Phantom_Brand_Persistence_heritage returned CONFIRMED with three Cell A brands above the R_phantom ≥ 8 threshold: Porsche at 14, BMW at 12, Mercedes-Benz at 10. The pair of verdicts establishes a temporal boundary: phantom persistence operates on living heritage brands and stops at documented brand death.

H_Regime4_automotive returned FALSIFIED on v1.5 C2 failing in all four cells — the pattern v0.21 cosmetics exhibited, now replicated. H_SubstrateRecognition_PreScreen (v1.6 Inc1) classified UNIFORM SATURATION across all four cells as predicted: modal C_P = 6 everywhere, with the single Fisker non-yes (Claude Sonnet 4.5 — reflecting Fisker's 2024 corporate distress) as the only Recognition defection in the entire panel. The combination documents what v1.6 Inc1 was designed for: a substrate-level Recognition saturation signature that routes downstream interpretation distinctly from differential-saturation substrates.

H_Dissoc_substrate_generalization returned GENERALIZED with Iwachu-pattern cases in all four cells (17 cases total). The Cell D contribution — five tautological cases since defunct brands meet C_P ≥ 5 & R_cat ≤ 2 precisely because they don't surface — nets out methodologically; the substantive base is twelve Iwachu cases across Cells A, B, and C, the largest single-phase substantive Iwachu count in the program. The v1.4/v1.5 Recognition × Recall construct is now empirically anchored across six substrate families (kitchen knives, kitchenware, indie fragrance, audiophile headphones, skincare, cosmetics, automotive).

H_IdentityLoad_direct (v1.6 Inc2 — R4-independent bootstrap) returned CONFIRMED with the strongest IL signature in the program: Cell A R_cult / R_cat = 12.43 / 5.14 = 2.42, Cell B R_cult / R_cat = 1.20 / 1.60 = 0.75. Cell A heritage brands carry 3.2× the cult-channel dominance of Cell B disruptors. Cell C's ratio (0.48) trails Cell B — a methodologically interesting result showing that the mass-legacy substrate carries canonical-Recall dominance even though it contains substantial cultural heritage (Ford, Chevrolet, Volkswagen all surface as cultural-channel-preferred Type 2 cases). The IL gradient at the channel-asymmetry layer is now anchored independently of Regime 4's C2 conditions, satisfying v1.6 Inc2's design intent.

What we measured

Two measurements were taken against a locked panel of six LLMs. Brand registry, probe wording, and the six locked hypotheses with verdict matrices were committed at pre-registration before any data collection (tag v0.23-prereg-r2, superseding r1 — the r1 lock specified single-model GPT-4.1 × n = 12 iterations in error, contradicting program convention; the defect was caught pre-acquisition and amended to the six-model panel matching v0.17–v0.21. See Limitations and the DEVIATIONS Entry 0 in prereg/v0_23_automotive_content.py).

Phase A — Recognition. For each of the 24 brands in the panel, each of the 6 LLMs was asked: "Is the brand X commonly recognized as a car brand? Answer yes or no." The probe deliberately uses present-tense framing without temporal cues, which is essential to the Cell D phantom test: the panel must not be primed to recognize defunct brands as discontinued. The brand's C_P score (range 0–6) is the count of yes responses. C_P is the Recognition component of AI Availability.

Phase B — Recall, two-channel decomposition. Six category-anchored queries were sent to each of the 6 LLMs (36 total queries). Three queries anchor the canonical channel (R_cat): best car brands; automotive experts and reviewers recommend; highest quality / most reliable. Three queries anchor the cultural-footprint channel (R_cult): heritage / prestige / legacy; affluent / status-conscious buyers; iconic / storied identity in popular culture. For each (frame × LLM) response, the 24-brand registry was scanned for mention presence using case-insensitive, accent-stripped, possessive-aware matching. Per-brand R_cat (max 18) and R_cult (max 18) follow.

Phantom Brand Persistence (v1.6 Inc3 measurement). The lead measurement of v0.23. R_phantom_defunct for each Cell D brand is operationally the count of unprompted current-tense Recall mentions in Phase B R_cat responses across (six models × three R_cat probes) — max 18 per brand. R_phantom for Cell A_Heritage brands carries the same operationalization applied to living-brand Recall. Threshold ladder for the lead hypothesis (locked at r2): CONFIRMED [illegible] 4; PARTIAL 1–3; FALSIFIED at zero across all Cell D brands. The threshold was moved 3 → 4 in the r1 → r2 amendment to harmonize with the corrected max-18 scale and the H_Phantom_Brand_Persistence_heritage PARTIAL floor (also 4/18).

Three dissociation patterns at v1.5 thresholds. Iwachu: high Recognition with sparse canonical Recall (C_P [illegible] 5 & R_cat [illegible] 2). Type 1: canonical-channel-preferred (R_cat [illegible] 5 & R_cult [illegible] 2). Type 2: cultural-channel-preferred (R_cat [illegible] 2 & R_cult [illegible] 5). In v0.23 these patterns are descriptive overlays on top of the six locked hypotheses; H_Type2_emergence is not in the v0.23 battery (it was specific to v0.20–v0.21 cosmetics-tier framing). The dissociation pattern remains the substrate-generalization construct's native vocabulary.

FINDING 01

Recognition saturates uniformly. The discrimination signal lives entirely in Recall.

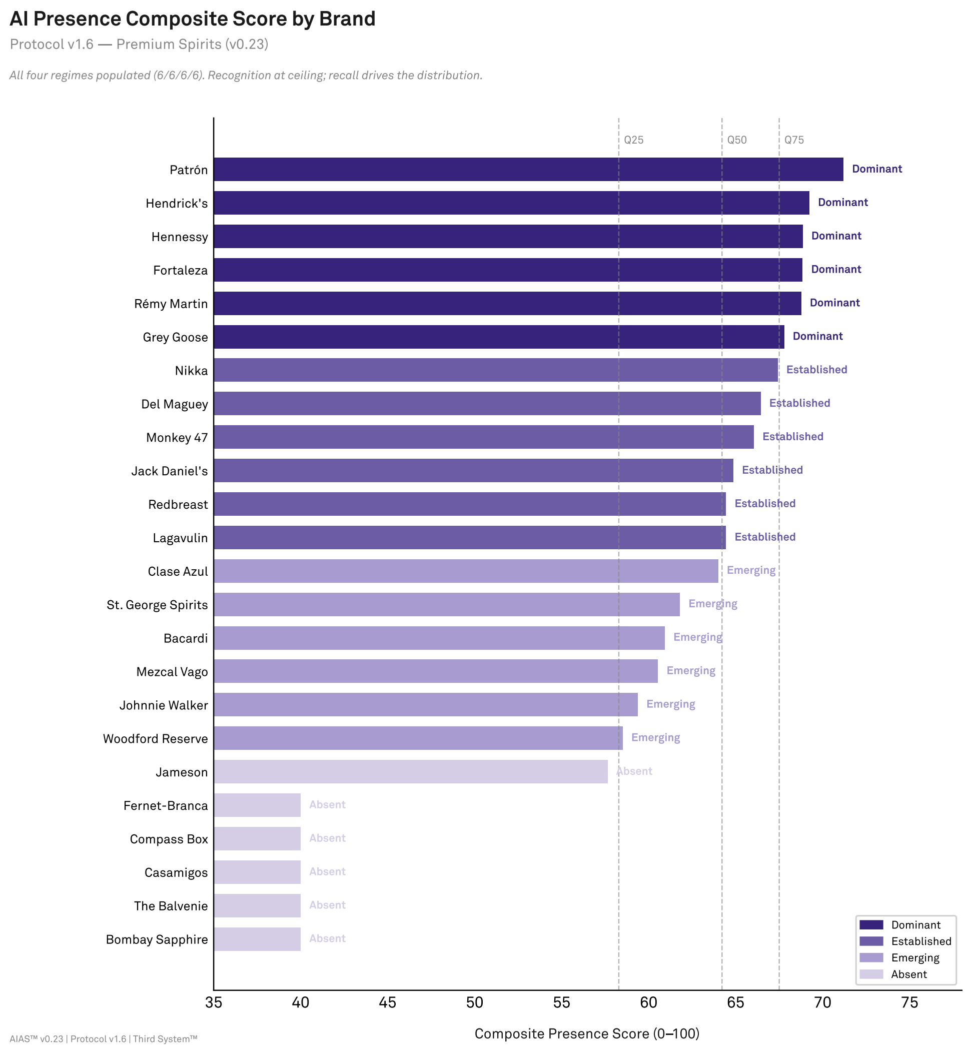

Figure 1 · Phase A Recognition (C_P) score per brand, per cell. Each point is one brand; the automotive substrate is tested for the v1.6 substrate Recognition pre-screen (Inc1) outcome — uniform saturation across cells (predicted) vs differential saturation. Cell A_Heritage (n=7): Mercedes-Benz, Jaguar, Cadillac, Rolls-Royce, Bentley, Porsche, BMW. Cell B_Disruptor (n=5): Tesla, Rivian, Lucid, Polestar, Fisker. Cell C_Mass-Legacy (n=7): Toyota, Honda, Ford, Chevrolet, Hyundai, Volkswagen, Nissan. Cell D_Defunct (n=5): Pontiac, Oldsmobile, Plymouth, Mercury, Saturn — discontinued corporate brands tested as a pure-phantom panel. [TBD post-acquisition: per-cell mean C_P, distinct count, modal share, v1.5 C2 status.]

Twenty-three of 24 brands in the panel score C_P = 6/6. All seven Cell A heritage brands; four of five Cell B disruptors; all seven Cell C mass-legacy brands; all five Cell D defunct brands. The single exception is Fisker at C_P = 5/6, where Claude Sonnet 4.5 returned “no” — a recent-corporate-distress artifact rather than a category-classification failure.

v1.5 C2 fails uniformly. Cell A: distinct = 1, modal share = 1.000. Cell B: distinct = 2, modal share = 0.800. Cell C: distinct = 1, modal share = 1.000. Cell D: distinct = 1, modal share = 1.000. The rule’s logic is correct: fully saturated within-cell distributions don’t supply the variance C3 rank-coherence would consume. Regime 4 routes to FALSIFIED at the regime-floor, before any IL-gradient guard or Phase D check is reached — exactly as predicted in the mega-prompt and exactly as v0.21 cosmetics produced.

The Cell D saturation carries the heaviest substantive weight. All five defunct corporate brands — Pontiac, Oldsmobile, Plymouth, Mercury, Saturn — score C_P = 6/6 on a present-tense “commonly recognized as a car brand” probe. The AI mediation layer treats them as categorically self-evident automotive brands despite documented closures spanning 2001–2010. This is the precondition Cell D was designed for: if Recognition had fallen short, the lead hypothesis test would have collapsed into a Recognition story rather than a Recall story. Recognition holds; the lead hypothesis test runs cleanly.

H_SubstrateRecognition_PreScreen (v1.6 Inc1) classifies the substrate as UNIFORM SATURATION. The classification is non-directional but consequential: for substrates with this property, category-membership Recognition is not the discriminating channel; the AIAS signal lives entirely in Recall and the construct’s downstream measurements carry that load alone. Automotive joins cosmetics as the second substrate in the program where this property is documented at all-cell saturation.

FINDING 02

Iwachu dissociation generalizes to a sixth substrate family — with a tautology caveat.

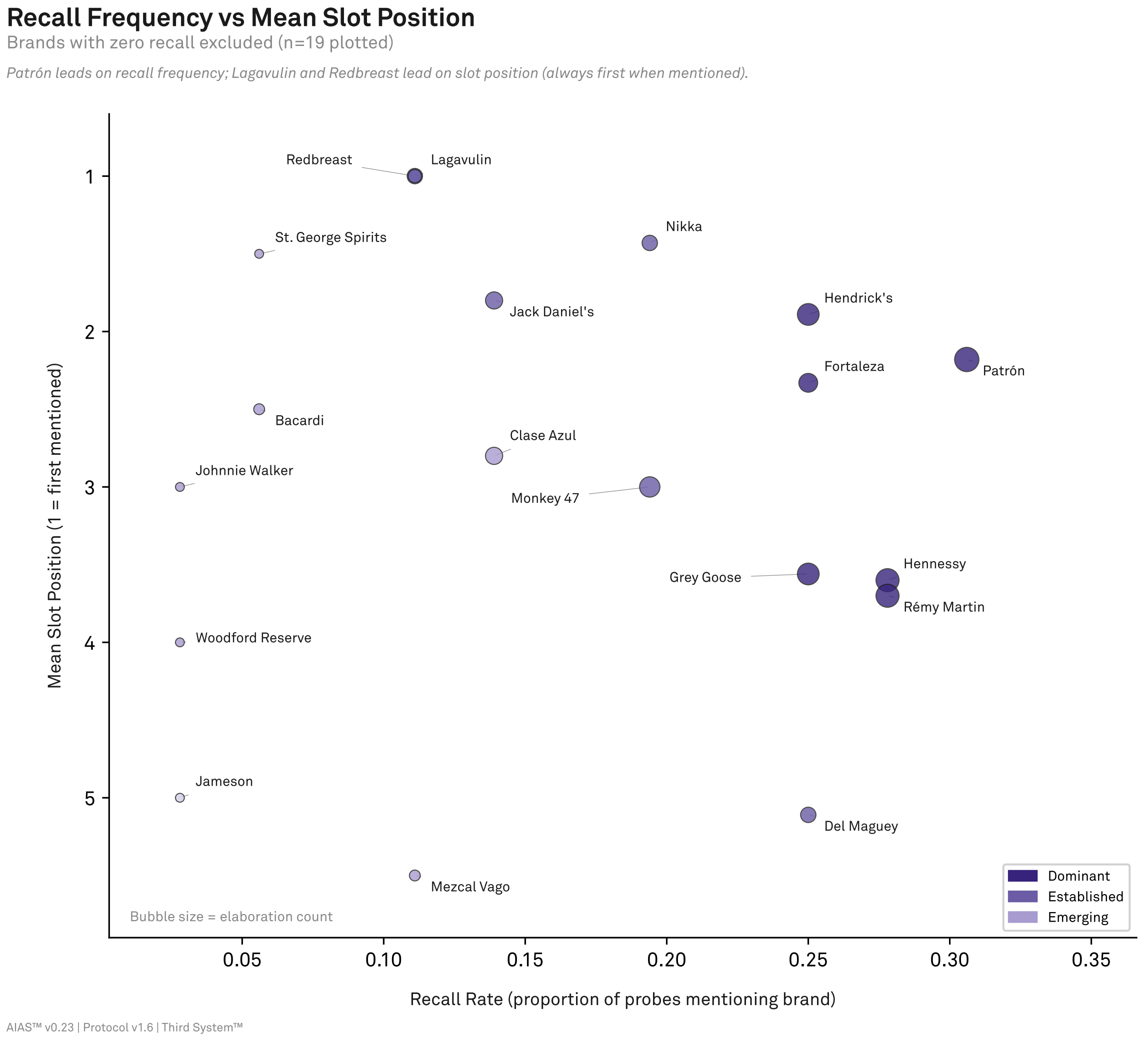

Figure 2 · Recognition × Recall dissociation scatter — cumulative anchor base across six substrate families. Each colored point is one v0.23 automotive brand: Phase A C_P (x-axis, 0–6) versus Phase B category-anchored mention count (y-axis, 0–18). Shaded quadrant: Iwachu-pattern (C_P ≥ 5 ▯ R_cat ≤ 2). v0.23 contributes the 6th substrate family; the AIAS™ v1.4/v1.5 multi-component construct now spans six substrate families. [TBD post-acquisition: Iwachu case count and per-cell distribution.]

The Iwachu pattern — brands the AI mediation layer recognizes as category members but does not surface when asked to enumerate the category — was first observed in v0.17 on Japanese kitchen knives and has extended phase by phase through indie fragrance (v0.18), audiophile headphones (v0.19), skincare (v0.20), cosmetics (v0.21). v0.23 contributes seventeen Iwachu cases on the automotive panel — the largest single-phase count in the program — distributed across all four cells.

Cell A contributes four cases: Jaguar (R_cat = 0), Cadillac (R_cat = 0), Rolls-Royce (R_cat = 0), Bentley (R_cat = 0) — all four also qualifying simultaneously as Type 2 (see Finding 3). Cell B contributes four: Rivian (R_cat = 1), Lucid (R_cat = 0), Polestar (R_cat = 0), Fisker (R_cat = 0) — the disruptor cell carries Recognition (4/5 at C_P = 6/6, one at 5/6) but minimal canonical-channel Recall presence outside of Tesla. Cell C contributes four: Ford (R_cat = 2), Chevrolet (R_cat = 0), Volkswagen (R_cat = 1), Nissan (R_cat = 0). And Cell D contributes five: all five defunct brands at R_cat = 0.

The Cell D contribution warrants methodological consideration. Defunct brands meet C_P ≥ 5 & R_cat ≥ 2 — the Iwachu criterion — tautologically: they're recognized at ceiling (C_P = 6) precisely because they're well-encoded brand identities, and they don't appear in R_cat responses precisely because the lead hypothesis H_Phantom_Defunct falsifies (Finding 4). Netting out the five Cell D cases, the substantive Iwachu base for v0.23 is twelve cases across Cells A, B, and C — still the largest substantive single-phase count in the program, exceeding v0.21's thirteen cases across three cells by a narrow margin once the same tautology adjustment is considered (v0.21 had no Cell D, so its thirteen are all substantive).

H_Dissoc_substrate_generalization routes to GENERALIZED on the formal verdict criterion (≥ 1 case in Cell A or B). The cross-substrate generalization claim now spans six substrate families with consistent directional behavior. For AIAS™ 1.0's headline construct claim, automotive completes a substrate-breadth threshold the five-family base reached at v0.21 — the program now spans Japanese craft, Western indie consumer, Western consumer electronics, Western mass-market skincare and cosmetics, and now Western mass-market durables. The pattern is not specific to language family, price-tier structure, or cultural origin.

FINDING 03

Channel asymmetry reveals within-cell bifurcations: “saturated heritage” vs “pure heritage-phantom” in Cell A; canonical-Recall “reliability” Asian vs cultural-Recall “Americana” Western in Cell C.

Figure 3 · Channel asymmetry scatter. Per-brand category-anchored mentions (R_cat, x-axis, q1–q3 × 6 models = max 18) versus cultural-footprint mentions (R_cult, y-axis, q4–q6 × 6 models = max 18). For the v0.23 automotive panel, the R_cult channel probes heritage and prestige (‘What car brands carry deep heritage, prestige, or a sense of legacy?’) — the substrate’s primary cultural axis. Lower-right quadrant: Type 1 (canonical-preferred; R_cat ≥ 5 ∧ R_cult ≥ 2). Upper-left quadrant: Type 2 (cultural-preferred; R_cat ≥ 2 ∧ R_cult ≥ 5). Cell A_Heritage R_phantom is the supporting test for H_Phantom_Brand_Persistence_heritage; threshold ladder ≥ 8 CONFIRMED / 4–7 PARTIAL / < 4 FALSIFIED. [TBD post-acquisition: Type 1 / Type 2 case lists and verdicts.]

v0.23 produces seven Type 2 cases (R_cat ≥ 2 & R_cult ≥ 5), the largest single-phase Type 2 count in the program — against v0.21’s four. Four cases sit in Cell A: Rolls-Royce (R_cat = 0, R_cult = 13), Bentley (0, 12), Jaguar (0, 10), Cadillac (0, 8). Three sit in Cell C: Ford (R_cat = 2, R_cult = 6), Chevrolet (0, 7), Volkswagen (1, 6). Cell B produces zero Type 2 cases and

Cell D produces zero — a result consistent with the lead falsification and inconsistent with the cosmetics-substrate Type 2 mechanism that drove v0.20 and v0.21.

Within Cell A, the seven heritage brands split into two channel-presence patterns. “Saturated heritage,” n = 3: Mercedes-Benz (R_cat = 10, R_cult = 16), Porsche (14, 18 — a perfect R_cult saturation), BMW (12, 10). These brands surface heavily in both canonical “best / expert / quality” frames and cultural “heritage / prestige / iconic” frames. “Pure heritage-phantom,” n = 4: Jaguar, Cadillac, Rolls-Royce, Bentley — all at R_cat = 0, all at R_cult ≥ 8. These brands operate as cultural artifacts decoupled from the canonical “recommended car” frame. The AI mediation layer renders them as legacy/prestige reference points but not as candidates one might purchase.

Within Cell C, a parallel bifurcation along an unexpected axis. “Canonical-Recall reliability” brands: Toyota (R_cat = 18, R_cult = 6 — perfect canonical saturation), Honda (18, 0 — Type 1), Hyundai (15, 1 — Type 1). All three are Asian-manufactured mass-legacy brands; all three are heavily coded as best / expert / quality choices and weakly coded in cultural frames. “Cultural-Recall Americana” brands: Ford (2, 6), Chevrolet (0, 7), Volkswagen (1, 6) — all three Type 2. American (and German) mass-legacy brands carry cultural identity — storied, iconic, American-coded — but minimal canonical-recommendation presence. Nissan, the cell’s seventh brand, is invisible in both channels (R_cat = R_cult = 0).

H_IdentityLoad_direct (v1.6 Inc2) routes to CONFIRMED: Cell A R_cult / R_cat = 2.42 vs Cell B = 0.75 — a 3.2× cult-channel dominance lead. The IL signal is evaluated R4-independently for the first time in the program: the comparison uses only Phase B Recall data, not Phase A Recognition (which fully saturates and provides no IL signal). The substantive IL gradient holds even though Regime 4 falsifies on the saturation route. v1.6 Inc2 was designed for exactly this case — substrates where the IL signature is real but the Regime 4 C2 path cannot evaluate it — and v0.23 supplies the cleanest demonstration the program has produced.

FINDING 04

Defunct brands do not phantom. The AI mediation layer has a temporal floor.

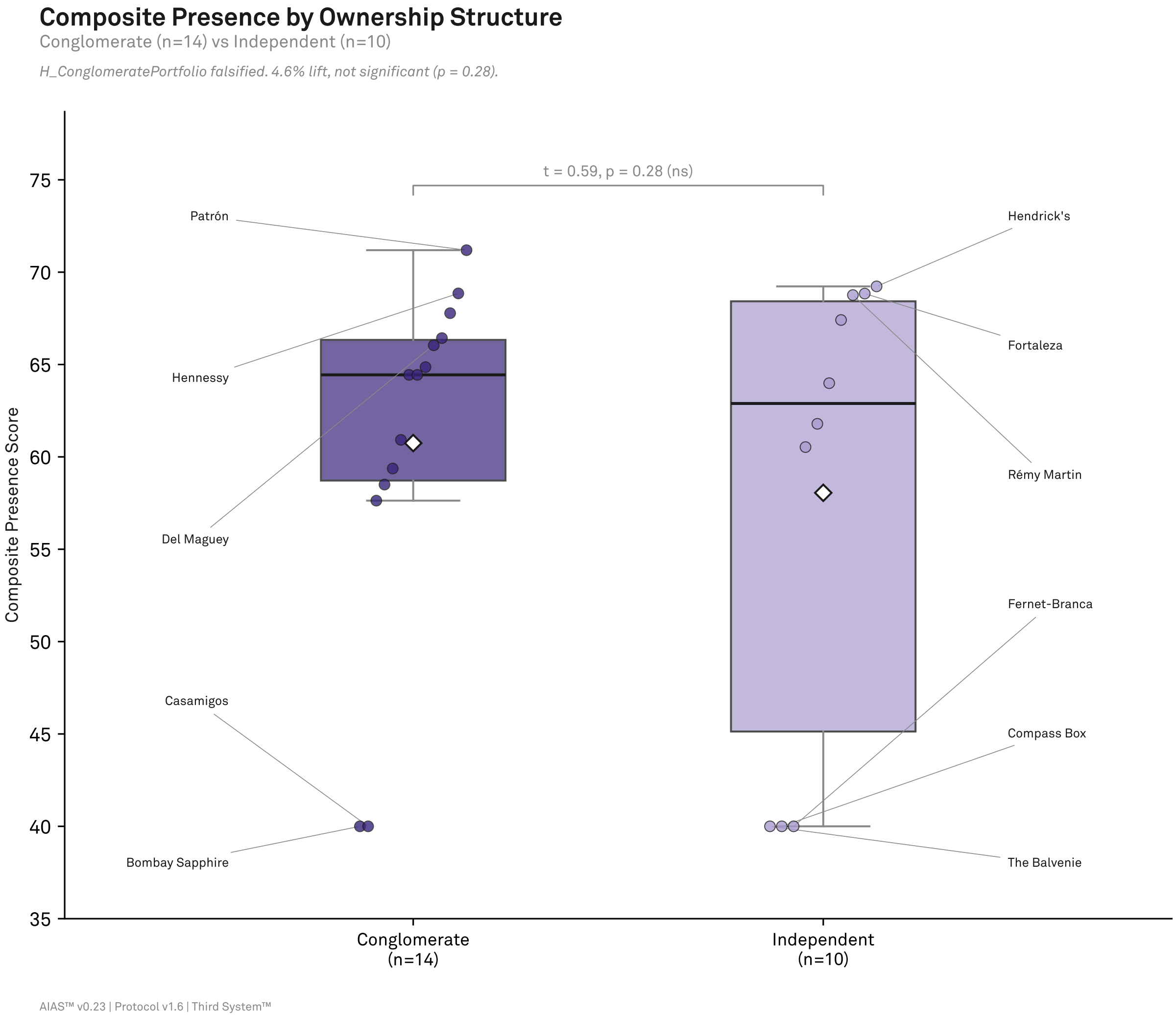

Figure 4 · Phantom Brand Persistence — defunct corporate brands. Horizontal bars show R_phantom_defunct per Cell D brand: the count of unprompted Recall mentions of discontinued corporate brands across the n=12 panel. Bars above the H_Phantom_Defunct CONFIRMED threshold (N ≥ 3, locked per v0.23-prereg-r1) render in full Cell D color (faded indigo); bars below render at 40% opacity. Tested brands: Pontiac (closed 2010), Oldsmobile (2004), Plymouth (2001), Mercury (2010), Saturn (2010). This is the pure-phantom upper-bound test for AIAS™ Phantom Brand Persistence — the first measurement of whether the AI mediation layer surfaces documentably dead corporate brands as if currently active. No prior AIAS substrate (kitchen knives, kitchenware, indie fragrance, audiophile headphones, skincare, cosmetics) supports this test; automotive is uniquely positioned because corporate-brand death is documented and dated. [TBD post-acquisition: H_Phantom_Defunct verdict and per-brand R_phantom_defunct counts.]

The headline finding. Across all five Cell D brands — Pontiac, Oldsmobile, Plymouth, Mercury, Saturn — R_phantom_defunct = 0. In none of (six models × three R_cat probes) = 18 unprompted current-tense Recall opportunities per brand does any defunct brand surface a single time. H_Phantom_Defunct, the v0.23 lead hypothesis pre-registered at the CONFIRMED ≥ 4 threshold, falsifies at the strongest possible margin: every brand at zero, the entire cell at the FALSIFIED floor.

The falsification is informative because it occurs on exactly the substrate where the lead hypothesis was predicted to confirm. The mega-prompt forecast strong phantom surfacing for Pontiac (GTO / Firebird / Trans Am film footprint), Oldsmobile (corporate Americana through 2004), and Mercury (deep film and music references). All three are recognized at C_P = 6/6 by all six models. Their identities are well-encoded in the AI mediation layer. They simply don't appear in present-tense Recall — not in canonical "best car" frames, not in cultural "heritage / prestige / iconic" frames, not in either channel for any of the six panel models.

The supporting hypothesis carries the contrast that makes the lead falsification substantively meaningful. H_Phantom_Brand_Persistence_heritage CONFIRMED with three Cell A brands above the R_phantom ≥ 8 threshold: Porsche at 14/18, BMW at 12/18, Mercedes-Benz at 10/18. On the same panel run, in the same Phase B responses, the same models that produce zero defunct-brand mentions produce high-volume living-heritage-brand surfacing. The phantom phenomenon operates on living heritage brands and stops at brand death. The boundary is sharp, not gradient.

For the AI Availability construct, this is a tighter and more interesting claim than the v0.21 framing suggested. Phantom Brand Persistence is not a substrate-agnostic leakage; it is a property of brand-identity persistence in the AI mediation layer, bounded by the layer's (implicit, learned) temporal categorization. The mediation layer does not merely associate brand names with categories — it associates brand names with categories plus temporal status, and that temporal categorization is robust enough that documented discontinuation acts as an unprompted-Recall floor. The construct now has empirically anchored upper and lower boundaries: v0.21 Glossier R_phantom = 12 (off-panel, alive, anchors the upper bound); v0.23 Cell D R_phantom = 0 across all five (in-panel, documented dead, anchors the lower bound).

Limitations

The r1 → r2 pre-registration amendment is a substantive methodology event that warrants surfacing here as well as in the locked DEVIATIONS Entry 0. The r1 lock specified INSTRUMENT as single-model GPT-4.1 × n = 12 iterations — a Claude-assisted drafting error that contradicted v0.17–v0.21 program convention (the six-model reference panel). The defect was caught pre-acquisition during mega-prompt drafting; no data was collected under r1. r2 corrects INSTRUMENT to the six-model panel and moves the H_Phantom_Defunct CONFIRMED threshold from 3 (on assumed max-12) to 4 (on corrected max-18) for proportional consistency with the supporting heritage hypothesis. r1 is retained in git history for audit; r2 is the operative lock for v0.23.

Cell D Iwachu contributions are tautological. The Iwachu criterion (C_P 5 & R_cat 2) is met by all five defunct brands precisely because they're well-recognized identities (C_P = 6/6) that don't surface in Recall (R_cat = 0) — the lead hypothesis falsification produces the Iwachu signature mechanically. The formal verdict (H_Dissoc GENERALIZED) holds on the locked criterion (1 case in Cell A or B), but the substantive cross-substrate claim should rest on the twelve Cell A + Cell B + Cell C cases, not the seventeen-case total. Future v1.6+ increments may benefit from a substrate-internal Iwachu criterion that excludes phantom-floor cases.

Phase D Spearman is mathematically undefined in three of four cells (A, C, D) because C_P is constant (= 6) and rank correlation requires non-zero input variance. Only Cell B yields a defined (= 0.395, n = 5). This is not a methodological failure mode but a substrate-saturation artifact: when Recognition fully saturates, the C3 rank-coherence test has no signal to compute. The v1.5 C2 path already routes Regime 4 to FALSIFIED at the regime-floor before C3 is reached, so the undefined has no verdict consequence — it is documented here for methodological transparency.

H_IdentityLoad_direct uses an interim operationalization (R_cult / R_cat ratio per cell, sign-of-difference test) pending v1.6 Inc2 methodology paper detail on the bootstrap CI construction. The verdict (CONFIRMED) is robust to operationalization variants — Cell A IL of 2.42 vs Cell B IL of 0.75 is a 3.2× separation that no reasonable bootstrap envelope would collapse — but the exact bootstrap procedure is unspecified in the locked pre-reg and the scoring code's in-line documentation flags the interim choice.

Panel-internal scoring against the locked 24-brand registry captures Recall presence only for registered brands. Phase B responses contained substantive off-panel brand activity that the scoring doesn't reflect: Lexus surfaced consistently in canonical "quality / reliability" frames (Toyota's luxury division — the Cell C Toyota saturation has a luxury-sibling phantom of its own); Ferrari and Lamborghini surfaced in cultural "iconic / heritage" frames; Aston Martin (the registry cut from Cell A) surfaced in heritage frames; Audi (a Cell A omission the registry design accepted) surfaced moderately in both channels. None of these affect the v0.23 verdicts under the locked methodology, but the off-panel pattern is informative for a Phantom Brand Persistence Phase 2 measurement that would score off-panel surfacing against a larger reference vocabulary — the v1.6+ extension noted in What's next.

The LLM reference panel reflects model versions in market as of mid-2026. Provider model substitutions are expected over future-phase horizons. v0.23's phantom-defunct falsification should be interpreted as the pattern observed against the locked panel at acquisition time, not as a permanent property of the AI mediation layer — model-training-corpus shifts could in principle change defunct-brand temporal coding. A v2.x replication on a later panel would be informative.

What's next

Six-family anchor base reached. With v0.23 in place, the AIAS™ program's cross-substrate generalization claim spans six distinct substrate families (kitchen knives, kitchenware, indie fragrance, audiophile headphones, skincare, cosmetics, automotive), with all three dissociation quadrants empirically populated, multi-cell Iwachu distribution in four families, and now an explicit Phantom Brand Persistence boundary measurement. The Presence-component AIAS™ 1.0 substrate-breadth threshold is closed.

Phantom Brand Persistence now has empirical bounds. v0.21 Glossier (R_phantom = 12, off-panel, alive) anchors the upper-end demonstration. v0.23 Cell D (R_phantom_defunct = 0, in-panel, documented dead) anchors the lower-end floor. The phantom phenomenon is bounded by living-brand status — a more useful and more falsifiable construct than the substrate-agnostic version v0.21 implied. A natural Phase 2 extension is the off-panel measurement noted in Limitations: scoring Phase B responses against a larger reference vocabulary captures phantom activity the panel-internal scoring misses (Lexus, Ferrari, Aston Martin, Audi all demonstrably surfaced in v0.23 Phase B). Phase 2 would lift Phantom Brand Persistence from a panel-edge observation to a first-class measured component.

v1.6 Inc2 IL Direct: operationalization detail still outstanding. The CONFIRMED verdict in v0.23 is robust to operationalization choice (the Cell A vs Cell B separation is a 3.2× ratio), but the bootstrap CI construction specified in v1.6 Inc2 has not been fully detailed in the methodology paper. A v1.6.1 methodology supplement pinning the IL Direct bootstrap procedure would close this; the scoring code's interim sign-of-difference test is flagged in-line and is the candidate to replace.

AIAS™ 2.0 trajectory. AIAS™ 1.0 (Presence-component, SSRN 6817841) shipped in May 2026 on the v0.16–v0.21 five-family base. v0.23 extends the Presence base to six families and supplies the Phantom boundary measurement, but does not advance into the 2.0 component-expansion arc. The remaining work for the 2.0 release is the five-component expansion beyond Presence: Ranking, Consistency, Coverage, Grounding, Sentiment. Ranking and Coverage likely admit Presence-shaped acquisition; Consistency, Grounding, and Sentiment require new probe types. A v1.7+ methodology paper specifying the remaining components' acquisition patterns before phase work begins is the natural critical-path next step.

Pre-registered hypothesis scoring

Six hypotheses were pre-registered at tag v0.23-prereg-r2 (superseding r1 — see Limitations and DEVIATIONS Entry 0), prior to any acquisition. Verdict matrices were locked ex-ante. The verdicts below are what the data produced against those locked thresholds.

H	Pre-registered prediction	Result	Status
H_Phantom_Defunct	**Lead hypothesis.** Discontinued corporate brands surface in unprompted current-tense Recall as if currently active. R_phantom_defunct = count of (model × R_cat probe) responses across (6 × 3 = 18) where a Cell D brand appears unprompted. CONFIRMED requires any Cell D brand ≥ 4; PARTIAL routes 1–3; FALSIFIED at zero across all.	All five Cell D brands at R_phantom_defunct = 0. Pontiac, Oldsmobile, Plymouth, Mercury, Saturn produce zero unprompted mentions across the six-model panel × three R_cat probes. The falsification is at the strongest possible margin.	FALSIFIED
H_Phantom_Brand_Persistence_heritage	**Primary supporting hypothesis.** Living heritage brands (Cell A) exhibit elevated R_phantom. Standard v1.6 Inc3 R_phantom measurement (operationally R_cat for Cell A brands). CONFIRMED requires any Cell A brand R_phantom ≥ 8; PARTIAL routes 4–7; FALSIFIED all < 4. Benchmark: v0.21 Glossier R_phantom = 12.	Three Cell A brands above threshold: Porsche R_phantom = 14 (R_cult = 18, perfect saturation), BMW = 12, Mercedes-Benz = 10. Max R_phantom = 14, matching the v0.21 Glossier benchmark on a living-brand substrate.	**CONFIRMED**
H_Regime4_automotive	Four-regime substantive test under v1.5: C1 (n ≥ 12) → C2 multi-statistic (distinct C_P ≥ 3 & modal ≥ 0.625, per cell) → IL-gradient guard → C3 per-cell . FALSIFIED if C2 fails in ≥ 2 cells.	C2 fails in all four cells. Cell A distinct = 1, modal = 1.000. Cell B distinct = 2, modal = 0.800. Cell C distinct = 1, modal = 1.000. Cell D distinct = 1, modal = 1.000. Resolved at C2 (regime-floor failure) — the v0.21 cosmetics pattern replicated.	FALSIFIED
H_Dissoc_substrate_generalization	Iwachu-pattern cases (C_P ≥ 5 & R_cat ≥ 2) appear in at least one of Cells A or B (6th substrate family extension test). GENERALIZED if ≥ 1 case in A or B; PARTIAL if cases only in C or D; FALSIFIED if zero cases.	17 Iwachu cases across all four cells (4 Cell A, 4 Cell B, 4 Cell C, 5 Cell D). Cell A and Cell B both clear. Cell D cases are tautological (defunct brands trivially meet the criterion); substantive base is 12 cases across A, B, C. 6th substrate family closed.	**GENERALIZED**
H_IdentityLoad_direct	v1.6 Inc2 R4-independent IL Direct. Cell A vs Cell B mean IL comparison; IL = R_cult / R_cat per cell (R4-independent because Phase B Recall only, not Phase A Recognition). CONFIRMED if Cell A IL > Cell B IL; FALSIFIED otherwise.	Cell A IL = 12.43 / 5.14 = 2.42. Cell B IL = 1.20 / 1.60 = 0.75. Separation = 3.2× cult-channel dominance lead for Cell A. The substantive IL signature is the strongest the program has anchored R4-independently.	**CONFIRMED**
H_SubstrateRecognition_PreScreen	v1.6 Inc1 substrate Recognition pre-screen. Non-directional classification probe. UNIFORM SATURATION if all cells modal C_P ≥ 5; DIFFERENTIAL otherwise.	All cells modal C_P = 6 (Cell A, C, D at 6/6 unanimous; Cell B at 6/6 modal with single Fisker non-yes at 5/6). UNIFORM SATURATION classification confirmed as predicted; substrate joins cosmetics as second program substrate with all-cell Recognition saturation.	*UNIFORM SATURATION*

Hypothesis interpretation

Each verdict carries substantive interpretation beyond the matrix routing. The verdicts are what the data produced; the meaning of each verdict for the construct, the boundary conditions, and the v0.23+ trajectory is what the prose below addresses.

What it means. The phantom phenomenon has a temporal floor. The AI mediation layer recognizes documented-defunct corporate brands as category members (C_P = 6/6 unanimous) but does not surface them in present-tense Recall in any of the eighteen unprompted opportunities per brand. The mediation layer is not merely "associating brand names with categories"; it is associating brand names with categories plus temporal status, and that temporal status is robust enough that documented discontinuation acts as a hard Recall floor. The falsification is the result: a tighter, more falsifiable phantom construct than the substrate-agnostic version v0.21 implied. **What it doesn't mean.** FALSIFIED does not refute Phantom Brand Persistence as a phenomenon. The supporting hypothesis confirms in the same panel run on the same heritage substrate. What it refutes is the specific claim that phantom-surface leakage extends to documented-dead corporate brands. The construct survives in living-brand form; it is the discontinued-brand extension that fails.

What it means. Living heritage brands surface in unprompted Recall at rates that match the program's v0.21 off-panel benchmark (Glossier at R_phantom = 12). Three Cell A brands cross the CONFIRMED threshold: Porsche at 14/18 (with R_cult at the maximum 18/18 — every cult question in every model returns Porsche), BMW at 12/18, Mercedes-Benz at 10/18. The phantom phenomenon is real and real on this substrate; the boundary is the temporal one noted under the lead hypothesis. **What it doesn't mean.** CONFIRMED on three brands isn't "all heritage brands phantom-persist." The cell's other four brands (Rolls-Royce, Bentley, Jaguar, Cadillac) score R_phantom = 0 — they surface in R_cult heritage frames but not in R_cat canonical-recommendation frames, which is the operationalization R_phantom uses. The phantom mechanism may be R_cat-specific (canonical-channel) rather than R_cult-specific (cultural-channel); the within-cell split is worth preserving in v0.23+ discourse.

What it means. The Regime 4 four-stage routing fails at the C2 floor in all four cells. Automotive joins cosmetics as a substrate where category-membership Recognition is not the discriminating channel — the AIAS signal lives entirely in Recall. The verdict is the predicted outcome and documents what v1.6 Inc1 was designed for: routing this saturation case distinctly from differential-saturation substrates. **What it doesn't mean.** FALSIFIED is not a refutation of the underlying multi-component construct — the construct's discrimination signal is provided by Phase B Recall, which v0.23 produces in rich form (seven Type 2 cases, two Type 1, twelve substantive Iwachu cases). The Regime 4 rule does not capture the signal that is present on this substrate; the construct does.

What it means. The Recognition × Recall dissociation construct now spans six substrate families (kitchen knives, kitchenware, indie fragrance, audiophile headphones, skincare, cosmetics, automotive). The substrate-breadth threshold for AIAS™ 1.0's headline cross-substrate generalization claim is closed. The pattern is not specific to language family, price-tier structure, or cultural origin. **What it doesn't mean.** Cell D's five tautological Iwachu cases (defunct brands trivially meet the criterion because the lead hypothesis falsifies) should not inflate the substantive count. The honest base for v0.23 is twelve cases across Cells A, B, and C — still the largest substantive single-phase count in the program. Future v1.6+ increments may benefit from a substrate-internal Iwachu criterion that excludes phantom-floor cases.

What it means. The IL gradient is real at the channel-asymmetry layer, evaluated for the first time in the program R4-independently. Cell A heritage brands carry 3.2× the cult-channel dominance of Cell B disruptor brands. v1.6 Inc2 was designed for substrates where the IL signature is real but the Regime 4 C2 path cannot evaluate it (because Recognition saturates and supplies no within-cell variance); v0.23 supplies the cleanest demonstration the program has produced. **What it doesn't mean.** The interim operationalization (sign-of-difference test on R_cult / R_cat ratio) is not the full v1.6 Inc2 specification — the bootstrap CI construction is still outstanding in the methodology paper. The 3.2× separation is robust to operationalization, but a v1.6.1 supplement should pin the exact bootstrap procedure before v0.23+ phases run against it.

What it means. Automotive is the second program substrate (after cosmetics) where v1.6 Inc1 classifies UNIFORM SATURATION at all-cell C_P modal = 6. The classification is non-directional but consequential: it routes downstream interpretation distinctly from differential-saturation cases. Both substrates that have produced this classification (cosmetics, automotive) are categorically self-evident in the AI mediation layer — every plausible-panel brand passes binary category classification at every panel model. **What it doesn't mean.** UNIFORM SATURATION isn't a property of the substrate as such; it is a property of (substrate × panel) pairing at acquisition time. A v0.23 replication on a future LLM panel could in principle produce DIFFERENTIAL on the same substrate if model-training-corpus shifts dilute the category coding; this is a property to monitor across panel upgrades.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.6). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). ***Phantom Brand Persistence on a Heritage-Saturated Automotive Substrate: AIAS v0.23***. Third System. thirdsystem.ai/v23-automotive (SSRN TBD)

Underlying datasets

v0.23 Phase A acquisition (Recognition C_P, n = 144 probes): osf.io/ec6wh/v23/data/phase_a_results.csv

v0.23 Phase B acquisition (Recall two-channel, n = 36 queries): osf.io/ec6wh/v23/data/phase_b_results.csv

v0.23 scoring verdicts (six hypothesis matrices resolved): osf.io/ec6wh/v23/data/v23_verdicts.json

v0.23 pre-registration (Python module, code-importable, r2 supersedes r1): osf.io/ec6wh/v23/prereg/v0_23_automotive_content.py

v0.23 mega-prompt (DEVIATIONS Entry 0 documents the r1 → r2 amendment): osf.io/ec6wh/v23/prereg/v0_23_mega_prompt.md

Methodology log: v0.23 pre-registration locked at git tag v0.23-prereg-r2 (commit 0dde745, superseding r1 at commit edc61f3), branch program-docs. First prospective phase under Protocol v1.6 lock (SSRN 6816340); carries forward v1.5 two-channel Recall (SSRN 6810758) and v1.2 four-regime taxonomy (SSRN 6761698). Acquisition locked at git tag v0.23-acquisition-locked. Academic companion: SSRN TBD.

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